



Physics-informed neural network and momentum contrastive learning for battery state of health estimation

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Abstract

Estimating the State of health (SoH) of lithium-ion batteries is essential for ensuring their safe and efficient operation across various applications. Traditional approaches often struggle to balance accuracy, physical consistency and data efficiency. This paper proposes a novel combination model of Physics-Informed Neural Network and Momentum Contrastive Learning for Battery State of Health Estimation that associates the interpretability of physics-based model with the representational power of contrastive learning. Our innovation lies in developing a unified optimization strategy that carefully balances an estimation physics-informed architecture and the power of contrastive learning. To specifically improve the physics-informed network, we leverage a shared feature encoder to improve representation learning for accurate SoH estimation. For contrastive learning, we design a physics-guided data augmentation strategy with a shared encoder, which generates realistic variations of battery degradation patterns and a momentum encoder architecture, which stabilizes the learning process. Extensive experiments on the NASA lithium-ion battery datasets demonstrate that our model achieves superior performance over state-of-the-art baselines such CNN, BPINN, Informer and XGBoost-ARIMA, achieving a mean absolute error (MAE) average of 0.095% and a root mean squared error (RMSE) average of 0.117% across all batteries. The associations of physics constraints with contrastive learning improve prediction accuracy and enhance model generalization across different battery types and operating conditions, addressing key limitations in existing battery health estimation approaches.

Keywords Lithium-ion batteries · State of health estimation · Physics-informed neural network · Contrastive learning · Battery management systems

Introduction

Lithium-ion Batteries (LIBs) have emerged as the dominant energy storage technology for portable electronic devices, electric vehicles, and renewable energy systems due to their high energy density, low self-discharge, and long cycle life [1–3]. However, LIBs inevitably experience capacity degradation over time due to complex electrochemical processes, including Solid Electrolyte Interphase (SEI) formation [4], lithium plating [5], and active material loss [6]. Therefore,

estimating battery State of Health (SoH) accurately is crucial for ensuring safe operation, optimizing charging protocols, and implementing effective maintenance strategies.

SoH directly reflects battery degradation and is a key indicator for battery management systems (BMS) [7]. Recent advances in deep learning have enabled more accurate SoH estimations by extracting complex patterns from battery operation data [8]. However, traditional approaches often lack interpretability and may violate physical laws, especially when training data is limited or contains noise. Physics-Informed Neural Network (PINN) [9] have emerged as a promising approach that incorporates physical laws into neural network training to improve estimation accuracy and ensure physical consistency [10]. Recent research highlights the need for more scalable, robust, and efficient algorithms in BMS, particularly as they integrate with the complex requirements of Autonomous Vehicles.

Despite these advancements, the existing PINN-based methods for SoH estimation still face critical limitations.

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First, their performance is heavily reliant on the availability of large, diverse, and well-labeled datasets, which are often scarce in battery applications. Second, their generalization capability across batteries with different degradation trajectories or under varying operational conditions remains a significant challenge. Most importantly, traditional PINN inherently lacks a mechanism to leverage the vast amounts of unlabeled battery data typically collected during operation. This underutilization of data prevents the model from learning more robust and invariant feature representations of the underlying degradation processes. These gaps highlight the need for a model that can enhance data efficiency and feature learning while maintaining physical consistency.

We propose a model that employs advanced physics-informed neural network and contrastive learning techniques for Battery State of Health Estimation. Our approach incorporates physics-guided data augmentation (PDA) to generate electrochemical variations of degradation patterns, which are then leveraged by a specially designed momentum-based contrastive learning mechanism working together with the Physics-Informed Neural Network-based Estimation (PINNE). This architecture simultaneously enforces physical consistency while extracting robust and generalizable feature representations that capture the fundamental patterns of battery degradation. By connecting physics-based modeling and contrastive learning, our architecture can effectively utilize both labeled and unlabeled data, leading to more accurate, physically consistent, and data-efficient SoH estimation. Our main contributions are summarized as follows:

- **A Novel Hybrid model:** We propose a new architecture that synergistically combines our adapted version of PINN for battery state of health estimation, referred to as Physics-informed Neural Network-based Estimation (PINNE), with a Momentum Contrastive Learning module, leveraging both labeled and unlabeled data for robust SoH estimation.
- **Physics-Guided Data Augmentation (PDA):** We design a novel PDA strategy that generates physically plausible variations of battery degradation patterns, providing meaningful negative pairs for contrastive learning and enhancing model robustness.
- **Advanced Contrastive Learning Mechanism:** We introduce a momentum encoder and an adaptive weighting strategy within our contrastive loss function to learn more discriminative and stable feature representations of battery health.

We demonstrate through extensive experiments on the NASA battery dataset that our model significantly outperforms existing methods in accuracy, data efficiency, and generalization across different batteries.

Related work

Battery Management Systems (BMS) are essential for maintaining the safety, longevity, and reliability of battery-powered applications such as EVs and grid storage systems [11–13]. Recent research has explored a variety of machine learning and deep learning techniques to enhance state-of-health (SoH) estimation and anomaly detection. This section reviews existing approaches across three key areas: battery health modeling, physics-informed neural network methods for time-series modeling, and contrastive learning, and highlights their limitations in BMS applications.

Data-driven approaches for battery health modeling

Data-driven approaches for battery SoH estimation have gained significant attention due to their ability to model complex nonlinear relationships without requiring detailed knowledge of battery internal chemistry. Early research relied on traditional machine learning techniques such as support vector machines (SVMs) [14] and random forests [15] to predict battery SoH based on features extracted from voltage, current, and temperature measurements. With the advancement of deep learning, more sophisticated neural network architectures have been applied to battery SoH estimation. Recurrent neural networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, have been widely used to capture the temporal dependencies in battery data [16–18]. Convolutional Neural Networks (CNNs) have also been employed to extract spatial features from battery data [17, 19, 20]. More recently, hybrid models such as a joint optimization approach combining XGBoost with ARIMA [21] to capture both non-linear feature relationships and temporal trends for SoH prediction have gained traction. Similarly, to address uncertainties in battery behavior a developed an improved quantile regression model [22] was developed using CNNs and RNNs, focusing on probabilistic battery temperature prediction, which is closely linked to SoH. While these methods show strong predictive power, they remain purely data-driven and do not explicitly enforce electrochemical principles.

Physics-informed neural networks for battery health modeling

Physics-based approaches model battery behavior using electrochemical principles and physical laws. These models provide better interpretability and can generate physically consistent predictions. Equivalent circuit models (ECMs) are widely used to represent a battery's electrical behavior using components such as resistors and capacitors [23–26]. More sophisticated electrochemical models, such as the pseudo-two-dimensional (P2D) model [27, 28], provide detailed descriptions of battery internal processes but

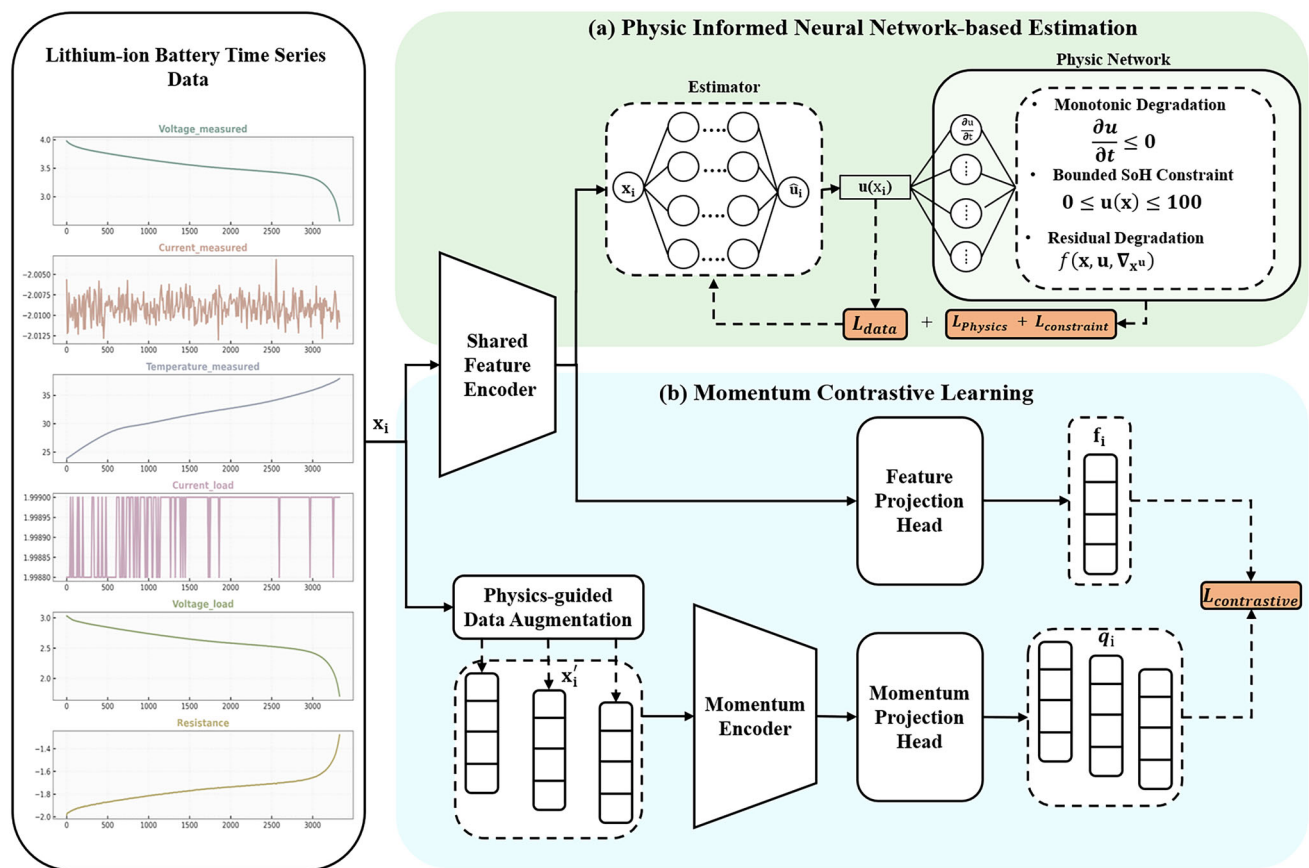


Fig. 1 Architecture of Physics-Informed Neural Network-based Estimation and Momentum Contrastive Learning for SoH estimation. (a) Physics-Informed Neural Network-based Estimation incorporating

degradation and constraints. (b) Momentum Contrastive Learning with physics-guided data augmentation, momentum encoder, projection heads, and queue-based memory bank

are computationally expensive. Recently, Physics-Informed Neural Networks (PINNs) [27, 29, 30] have emerged as a promising approach for battery SoH prediction. PINNs integrate physical laws into neural network training by adding physics-based regularization terms to the loss function [27]. While PINN based methods have been used for battery SoH prediction, Wei et al. [27] investigated a multilevel data-driven battery management approach that combines physical modeling with big data analytics to improve battery monitoring and decision making.

To balance accuracy and computational efficiency, hybrid approaches that combine physics-based models with data-driven techniques associated with contrastive learning can be more efficient. When applied in this context, contrastive learning enhances the model's ability to distinguish subtle differences in degradation patterns by maximizing agreement between physically meaningful augmented views of the same battery time series data samples and minimizing agreement between different samples.

Contrastive learning for battery health modeling

Contrastive learning is a self-supervised learning technique that learns data representations by maximizing agreement between differently augmented views of the same sample (positive pairs) while minimizing agreement between views of different samples (negative pairs) [31–33]. This approach has achieved remarkable success in computer vision and natural language processing, particularly in scenarios with limited labeled data. Several contrastive learning frameworks have been proposed, including SimCLR [31], MoCo [32], and BYOL [34], each with different strategies for generating positive and negative pairs and computing contrastive loss. These approaches have been primarily applied to image and text data, with limited exploration in time-series domains like battery data. A few recent studies have begun to explore contrastive learning for time-series analysis. Eldele et al. [35] proposed a time-series representation learning framework based on contrastive learning that outperforms supervised approaches in various classification tasks. Franceschi et al. [36] introduced a contrastive learning approach for multivari-

ate time series that achieves state-of-the-art results on several benchmark datasets.

However, the application of contrastive learning to battery SoH estimation remains largely unexplored, particularly in combination with physics-informed approaches. Our work bridges this gap by associating contrastive learning with PINN to enhance battery SoH estimation.

Method

We propose a model for Battery State of Health Estimation through advanced Physics-Informed Neural Network and Momentum Contrastive Learning, illustrated in Fig. 1, to enforce physical constraints on battery electrochemical degradation and to capture intrinsic patterns in battery time series data. Our model is constituted by an Physics-Informed Neural Network-based Estimation (PINNE) component and a Momentum Contrastive Learning component that is complemented by a physics-guided data augmentation strategy that generates physically and electrochemically accurate variations of battery degradation. These two components are connected through a shared feature encoder, forming a jointly trained unified architecture. This unified architecture ensures electrochemical consistency and robust feature representations, enabling accurate estimation of battery state of health (SoH) across diverse operating conditions.

For battery State of Health (SoH) estimation, we utilize a set of features collected from a battery during operations. These features capture various physical and electrochemical aspects influencing battery degradation. Table 1 summarizes the set of features and their corresponding notations used throughout this work. Given battery time series data $\mathbf{x}_i = \{\mathbf{v}_i, \mathbf{c}_i, \mathbf{o}_i, \mathbf{t}_i, \mathbf{s}_i, \mathbf{r}_i, \mathbf{p}_i, \mathbf{q}_i, \mathbf{w}_i\}$ collected during battery operation at i th step, we define $\mathbf{n}_i$ as a subset of $\mathbf{x}_i$ with $\mathbf{n}_i = \{\mathbf{v}_i, \mathbf{c}_i, \mathbf{o}_i, \mathbf{s}_i, \mathbf{r}_i, \mathbf{p}_i, \mathbf{q}_i, \mathbf{w}_i\}$ representing all features excluding time feature $\mathbf{t}_i$ that will be encoded separately. The goal is to estimate battery State of Health (SoH) value $\hat{\mathbf{u}}_i \in [0, 100]$.

Physics-informed neural network-based estimation

The Physics-Informed Neural Network-based Estimation (PINNE) component of our model consists of a shared feature encoder that is used in momentum contrastive learning component, an estimator that estimates SoH values directly from feature embeddings produced by the shared feature encoder, and a physics network that enforces physical constraints on estimated SoH values.

First, the shared feature encoder in Fig. 2 passes the battery time series data through successive linear layers, layer normalization, and Sigmoid Linear Unit (SiLU) activation [37] which multiplies each input by a sigmoid function. For-

Table 1 Battery features and their symbols

fFeature	Symbol
Voltage	v
Current	c
Temperature	o
Time	t
State of Charge (SoC)	s
Internal Resistance	r
Capacity	p
Current Load	q
Voltage Load	w

mally, for input features, $\mathbf{n}_i$ and $\mathbf{t}_i$, the shared feature encoder outputs the embeddings, $\mathbf{z}_i$ and $\mathbf{h}_i$.

Second, the estimator receives concatenated features embeddings, $\mathbf{z}_i$ and $\mathbf{h}_i$, and outputs an initial estimated SoH value. In the last layer of the estimator, an explicit time-dependent term is added to the estimated SoH value $\hat{\mathbf{u}}_i$:

$$\hat{\mathbf{u}}_i = \text{Estimator}([\mathbf{z}_i, \mathbf{h}_i]) + \sin(\pi \times \mathbf{t}_i) + \cos(2\pi \times \mathbf{t}_i) \quad (1)$$

The addition of explicit time-dependent sinusoidal terms is used in Fourier Feature Mapping and Positional Encoding, designed to enhance the network's ability to model complex temporal dynamics [38, 39]. By mapping the low-dimensional time input t_i into a higher-dimensional feature space using periodic sine and cosine functions, the network can more easily approximate complex, high-frequency patterns within the battery degradation trajectory. This approach is not intended to model the non-periodic SoH degradation as a periodic function, but rather to provide the network with a richer set of basis functions to represent the signal. Furthermore, as periodic functions have non-zero gradients across their domain, this formulation also improves the gradient flow during backpropagation, contributing to more stable and efficient training [40].

Finally, our physics network $\zeta(\cdot)$ ensures that estimated SoH values satisfy physical laws governing battery degradation. While the estimator provides a step-wise estimation $\hat{\mathbf{u}}_i$ of SoH value, the physics network treats SoH values as a function of the battery features $\mathbf{u}(\mathbf{x})$, enabling the application of differential operators and the enforcement of physics-informed constraints. It is represented as $\mathbf{u}$ in the following equations for simplicity:

- **Monotonic Degradation constraint:** This constraint is based on the fundamental principle that, under normal operating conditions, a battery's capacity is non-increasing due to irreversible electrochemical and mechanical degradation processes such as SEI growth and active

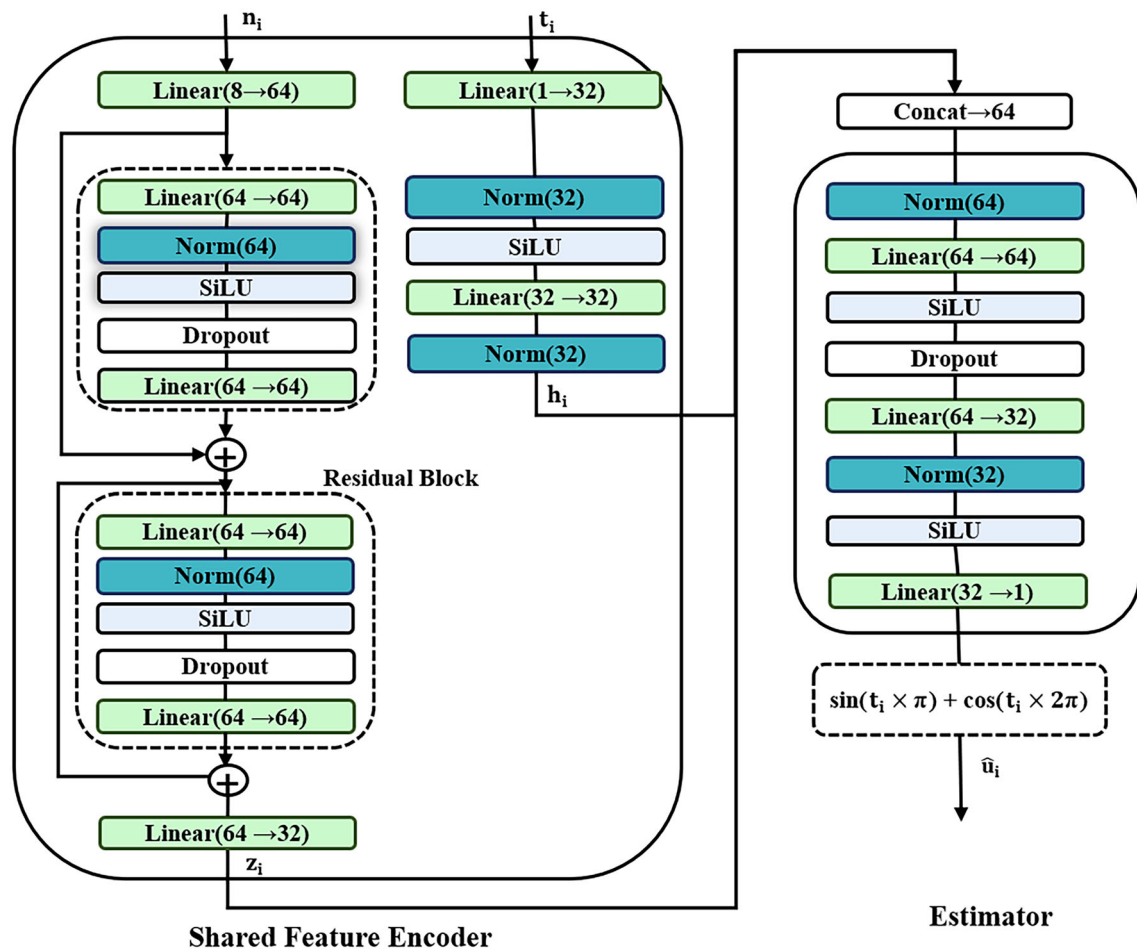


Fig. 2 Architecture of the shared feature encoder and estimator

material loss [41]. Therefore, the SoH, which is a direct measure of this capacity, must decrease or remain constant over time:

$$\frac{\partial \mathbf{u}}{\partial t} \leq 0 \quad (2)$$

- **Bounded SoH Values constraint:** The State of Health is defined as a percentage of the nominal capacity, ranging from 100% (a new battery) to 0% (a completely failed battery). This constraint enforces the physical definition of SoH, ensuring that the model's predictions remain within a meaningful and valid range:

$$0 \leq \mathbf{u} \leq 100 \quad (3)$$

- **Battery Degradation Dynamics:** We approximate the overall degradation dynamics using semi-empirical models that capture the most dominant aging mechanisms. Our physics network approximates the ideal physical function $E(\mathbf{x}, \mathbf{u}, \nabla_{\mathbf{x}} \mathbf{u})$ by incorporating three well-established degradation models: SEI growth, calendar

aging, and cycling degradation:

$$\frac{\partial \mathbf{u}_{\text{ideal}}}{\partial t} \rightarrow E(\mathbf{x}, \mathbf{u}, \nabla_{\mathbf{x}} \mathbf{u}) \quad (4)$$

where $E(\mathbf{x}, \mathbf{u}, \nabla_{\mathbf{x}} \mathbf{u})$ represents an underlying ideal electrochemical dynamics degradation and $\nabla_{\mathbf{x}} \mathbf{u}$ represents the gradient of SoH with respect to all feature variables:

To better approximate $E(\mathbf{x}, \mathbf{u}, \nabla_{\mathbf{x}} \mathbf{u})$, we incorporate known physics-informed battery degradation mechanisms

1. **SEI Growth** [42]: The growth of the Solid Electrolyte Interphase (SEI) layer is a primary cause of capacity fade, driven by the diffusion of electrolyte solvent molecules through the existing SEI layer to the anode surface. As the SEI layer thickens, it acts as a barrier, slowing down the diffusion rate. This diffusion-limited process is widely modeled by a square-root time law, where the rate of degradation is inversely proportional to the square root of time [43]. We model this fundamental mechanism as

follows:

$$\frac{\partial \mathbf{u}_{\text{sei}}}{\partial t} = -\frac{k_{\text{sei}}}{2\sqrt{t} + \epsilon} \quad (5)$$

where $\mathbf{u}_{\text{sei}}$ is the SoH for SEI Growth, k_{sei} is the SEI growth rate constant and ϵ is a small constant to avoid division by zero:

2. **Calendar Aging** [44]: Calendar aging refers to degradation that occurs even when the battery is at rest, driven by parasitic chemical reactions. The rates of these reactions are highly dependent on temperature. The Arrhenius equation is the fundamental model used in chemistry to describe this temperature dependency, linking the reaction rate constant to temperature and an activation energy (E_a) [45, 46]. We incorporate this principle to model the acceleration of aging at higher temperatures:

$$\frac{\partial \mathbf{u}_{\text{cal}}}{\partial t} = -A \cdot \exp\left(\frac{-E_a}{R \cdot \mathbf{o}}\right) \cdot \frac{1}{\sqrt{t} + \epsilon} \quad (6)$$

where $\mathbf{u}_{\text{cal}}$ is the SoH for Calendar Aging, A is a fixed pre-exponential factor, E_a is a normalized activation energy, R is the gas constant, and $\mathbf{o}$ is a temperature feature.

3. **Cycling Degradation** [47]: In addition to calendar aging, degradation is accelerated by the stresses of charging and discharging. This cycling degradation is heavily influenced by operational conditions, particularly the State of Charge (SoC). It is well-documented that battery degradation is more pronounced at both very high and very low SoC levels [48]. We model this using a stress function $f_{\text{stress}}(s)$ that captures the U-shaped dependency on SoC, combined with the absolute current $|c|$:

$$\frac{\partial \mathbf{u}_{\text{cyc}}}{\partial t} = -\mathbf{B} \cdot f_{\text{stress}}(s) \cdot |c| \quad (7)$$

where $\mathbf{u}_{\text{cal}}$ is the SoH for Cycling Degradation, $\mathbf{B}$ is a fixed cycling degradation factor, $f_{\text{stress}}(s) = 4\left(\frac{s}{100} - 0.5\right)^2$. In this equation s is the state of charge and $|c|$ is the absolute value of current.

The final physics-informed degradation rate is the sum of all three mechanisms:

$$\frac{\partial \mathbf{u}_{\text{ideal}}}{\partial t} = \frac{\partial \mathbf{u}_{\text{sei}}}{\partial t} + \frac{\partial \mathbf{u}_{\text{cal}}}{\partial t} + \frac{\partial \mathbf{u}_{\text{cyc}}}{\partial t} \quad (8)$$

The physics network ζ learns to approximate the residual difference between the predicted $\mathbf{u}$ and the ideal degradation rate governed by the ideal physical function $\mathbf{E}(\cdot)$:

$$\zeta(\mathbf{x}, \mathbf{u}, \nabla_{\mathbf{x}} \mathbf{u}) \approx \frac{\partial \mathbf{u}}{\partial t} - \mathbf{E}(\mathbf{x}, \mathbf{u}, \nabla_{\mathbf{x}} \mathbf{u}) \quad (9)$$

We minimize this residual during training to enforce physical consistency.

The overall PINNE loss is:

$$\mathcal{L}_{\text{data}} = \frac{1}{N} \sum_{i=1}^N (\hat{\mathbf{u}}_i - \mathbf{u}_i)^2 \quad (10)$$

$$\mathcal{L}_{\text{physics}} = \frac{1}{N} \sum_{i=1}^N (\zeta(\mathbf{x}_i, \mathbf{u}_i, \nabla_{\mathbf{x}} \mathbf{u}_i))^2 \quad (11)$$

$$\mathcal{L}_{\text{constraint}} = \frac{1}{N} \sum_{i=1}^N \left(\text{ReLU}\left(\frac{\partial \mathbf{u}_i}{\partial \mathbf{x}_i}\right) + \text{ReLU}(-\mathbf{u}_i) + \text{ReLU}(\mathbf{u}_i - 100) \right) \quad (12)$$

$$\mathcal{L}_{\text{PINNE}} = \mathcal{L}_{\text{data}} + \alpha \mathcal{L}_{\text{physics}} + \beta \mathcal{L}_{\text{constraint}} \quad (13)$$

where $\mathbf{u}_i$ is the ground truth(true), $\hat{\mathbf{u}}_i$ is the predicted value, and α and β are weighting parameters that balance respectively the importance of physics and constraint. and N is batch of a battery time series data.

Momentum contrastive learning

To enhance the feature representations learned by our PINNE, we introduce a momentum contrastive learning module that leverages unlabeled battery data through Physics-guided Data Augmentation(PDA) and Momentum Encoder. The key idea is to learn representations that capture intrinsic battery degradation patterns by maximizing similarity between augmented and original views at i th step of battery time series data features and minimizing similarity between different steps.

First, to improve the robustness of our model and enable effective contrastive learning, we propose a physics-guided data augmentation (PDA) strategy that generates realistic variations of battery degradation. These augmentations are based on physical processes underlying battery degradation and preserve physical plausibility. The PDA strategy consists of the following key augmentation mechanisms:

1. **SEI Growth Augmentation:** The formation and growth of the Solid Electrolyte Interphase (SEI) layer on the battery anode is a dominant contributor to capacity fade, especially in the early cycles. To simulate this effect, we perturb the State of Health (SoH):

$$\mathbf{u}'_i = \mathbf{u}_i - k \sqrt{t_i} \quad (14)$$

where t_i denotes normalized cycle time, and k is a SEI growth constant sampled from a physically reasonable range for each instance.

2. **Resistance Increase Augmentation:** Progressive increase in internal resistance is a hallmark of battery aging. We

model resistance growth by incrementing the original resistance value:

$$\mathbf{r}'_i = \mathbf{r}_i + \Delta \mathbf{r}_i \quad (15)$$

$$\mathbf{v}'_i = \mathbf{v}_i - \Delta \mathbf{r}_i \cdot |\mathbf{c}_i| \quad (16)$$

3. **Temperature Effects Augmentation:** Temperature exerts a strong influence on battery performance and degradation rates. The temperature dependence of degradation is described by the Arrhenius equation:

$$G = K \cdot \exp\left(\frac{-E_a}{R} \left(\frac{1}{o_i} - \frac{1}{o_1}\right)\right) \quad (17)$$

where $\mathbf{G}$ is the reaction rate, $\mathbf{K}$ is the pre-exponential factor, $\mathbf{E}_a$ is the activation energy, $\mathbf{R}$ is the universal gas constant, $\mathbf{o}_i$ is the temperature, and $\mathbf{o}_1$ is the initial reference temperature.

4. **Cycle Aging Augmentation:** Cycle aging refers to the loss of capacity attributable to repetitive charge-discharge cycles, which depends strongly on the depth of discharge D_i and the time. To model this, we perturb SoH as:

$$D_i = 1 - \frac{s_i}{100} \quad (18)$$

$$s'_i = s_i - \lambda \cdot (D_i)^2 \cdot t_i \quad (19)$$

where λ is randomly sampled from a uniform distribution in the range $[0.0001, 0.001]$ and t_i is the time feature.

Finally, our momentum contrastive learning leverages a shared feature encoder $\mathbf{e}_{\phi_f}(\cdot)$ also used by our Physics Informed Neural Network-based Estimation component, a momentum encoder $\mathbf{e}_{\phi_q}(\cdot)$ that is a slowly-updating copy of the shared feature encoder, a momentum projection head $\mathbf{g}_{\phi_q}(\cdot)$, and a feature projection head $\mathbf{g}_{\phi_f}(\cdot)$ to encode the augmented and the original battery time series data, respectively, into the same embedding representation space.

The feature projection head $\mathbf{g}_{\psi_f}(\cdot)$ and momentum projection head $\mathbf{g}_{\psi_q}(\cdot)$ are small neural networks, typically Multi-Layer Perceptrons (MLPs) with a projection dimension of 128, that map the representations from the encoders to a space where the contrastive loss is applied [31]. This architectural design serves a crucial purpose: it allows the encoder $\mathbf{e}_{\phi_f}(\cdot)$ to learn representations that are rich in information for the downstream task (SoH estimation), while the projection heads specialize in distilling features specifically optimized for the contrastive learning objective [49]. This separation prevents the contrastive loss from directly constraining the encoder's representation space, thereby preserving the richness of features needed for accurate SoH prediction. The momentum encoder and momentum projection head update their parameters ϕ^q and ψ^q using a moving average of the

shared feature encoder parameter ϕ^f and feature projection head ψ^q for stable parameters updates:

$$\phi^q \leftarrow m \cdot \phi^q + (1 - m) \cdot \phi^f \quad (20)$$

$$\psi^q \leftarrow m \cdot \psi^q + (1 - m) \cdot \psi^f \quad (21)$$

where m is the momentum coefficient.

This momentum update mechanism is a core component of MoCo [32]. Critically, the momentum encoder's parameters ϕ_q are not updated by back-propagation through the contrastive loss. Instead, they evolve as an exponential moving average of the query encoder's parameters ϕ_f . This ensures that the momentum encoder changes slowly and smoothly over training iterations. This slow evolution is essential for three key reasons. First, it provides consistency in the negative sample dictionary. Since we maintain a queue of encoded negative samples from previous mini-batches, these samples were encoded by earlier versions of the encoder. If the encoder parameters changed rapidly, the encoded representations in the queue would become inconsistent with newly encoded samples, destabilizing the contrastive learning signal. The momentum mechanism ensures all samples in the queue remain comparable. Second, it enables stable gradient flow. Rapidly changing encoder parameters can lead to volatile contrastive losses, where the definition of positive and negative pairs shifts dramatically between iterations. The momentum encoder acts as a stabilizing anchor, smoothing these fluctuations. Third, it facilitates learning from a larger effective batch size. By maintaining consistency across mini-batches, the model effectively contrasts each sample against a much larger set of negatives than would fit in a single mini-batch, improving representation quality without requiring prohibitively large memory.

The shared encoder architecture creates a powerful synergy between Physics-Informed Neural Network-based Estimation (PINNE) and contrastive learning components. During training, the shared encoder $\mathbf{e}_{\phi_f}(\cdot)$ receives gradient signals from the physics-informed loss, which encourages representations that respect electrochemical degradation laws, then from the contrastive loss. This dual supervision enables the encoder to learn features that are simultaneously physically consistent and robustly generalizable. The momentum encoder, by maintaining a slowly-updating version of these features, provides a stable target for contrastive learning that reflects the accumulated knowledge of battery degradation patterns learned throughout training.

Given a batch of N battery time series data measurements, we generate an augmented view of each sample through physics-guided augmentations (PDA). For a given sample $\mathbf{x}_i$, we denote its augmented counterpart as $\mathbf{x}'_i$, while

all other samples in the batch serve as potential negatives. The embeddings produced by the feature projection head and momentum projection head are respectively denoted as $f_i = g_{\psi f}(e_{\phi f}(\mathbf{x}_i))$ and $q_i = g_{\psi q}(e_{\phi q}(\mathbf{x}'_i))$.

Our contrastive loss function is defined as:

$$\mathcal{L}_{\text{contrastive}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(f_i \cdot \frac{q_i}{\tau})}{\exp(f_i \cdot \frac{q_i}{\tau}) + \sum_{\substack{j=1 \\ j \neq i}}^N w_{ij} \exp(f_i \cdot \frac{f_j}{\tau})} \quad (22)$$

where τ is the temperature parameter and w_{ij} is an adaptive weighting factor based on SoH similarity and steps j over all negative samples (other data steps not equal to i).

The adaptive weighting factor w_{ij} is defined as:

$$w_{ij} = \begin{cases} 1.0, & \text{if } |u_i - u_j| > \delta \\ \alpha \cdot \frac{|u_i - u_j|}{\delta}, & \text{otherwise} \end{cases} \quad (23)$$

where u_i and u_j are the SoH values of samples at steps i and j , δ is a SoH similarity threshold, and α is a fixed scaling factor.

The adaptive weighting factor w_{ij} is designed to handle hard negative samples, which are temporally distinct from the anchor sample i but have a similar SoH value. Standard contrastive loss treats all negative samples equally, but samples with similar SoH values can confuse the model since they represent similar degradation states despite being from different time steps. Our weighting scheme reduces the penalty for these hard negatives, making the training more robust. As mentioned in our experiments, the hyperparameters were set based on empirical validation. We used a SoH similarity threshold of $\delta = 5.0$ to distinguish between significantly different and similar degradation states. The scaling factor was set to $\alpha = 0.5$ to moderately reduce the influence of hard negative samples. These values were chosen to balance the trade-off between learning discriminative features and avoiding instability caused by potentially misleading hard negatives.

During training, we maintain a queue of previously encoded samples following the MoCo approach [32]. This queue serves as a dynamic dictionary of negative samples, allowing the model to contrast against a large and consistent set of features from preceding mini-batches. This design decouples the number of negative samples from the mini-batch size, which is a key advantage over methods like SimCLR that only use negatives from the current batch. A larger queue generally leads to better performance as it provides a more diverse sampling of the data distribution, enabling the model to learn more discriminative features.

Optimization strategy

The proposed model is formulated as a Unified optimization problem that incorporates both Physics-Informed Neural Network-based Estimation (PINNE) objectives and momentum contrastive learning objectives. The overall loss function is defined as:

$$\mathcal{L}_{\text{final}} = \mathcal{L}_{\text{PINNE}} + \gamma \mathcal{L}_{\text{contrastive}} \quad (24)$$

where $\mathcal{L}_{\text{PINNE}}$ represents the physics-informed loss, and $\mathcal{L}_{\text{contrastive}}$ the contrastive learning loss. The parameter γ is adaptively adjusted during training:

$$\gamma = \gamma_0 \cdot \min\left(1.0, \frac{\mathcal{L}_{\text{PINNE}}}{\mathcal{L}_{\text{contrastive}} + \epsilon}\right) \quad (25)$$

where γ_0 is the initial contrastive loss weight and ϵ is a small constant for numerical stability.

This optimization scheme ensures that the model simultaneously learns to satisfy the physical laws governing battery degradation and to extract robust, generalizable feature representations via contrastive learning.

Experiments setup

Implementation details

All experiments were conducted on a server equipped with a CPU of 32 GB RAM, and a single NVIDIA GeForce RTX 4090 GPU with 24 GB of VRAM. Our proposed model is computationally efficient, with a total of approximately 1.96 million trainable parameters (131k for the PINNE and 1.83M for the momentum contrastive learning). The shared feature encoder and the momentum encoder consisted of 8 layers with a hidden dimension of 128. The model was trained for 500 epochs with a batch size of 512. We used an Adam optimizer with an initial learning rate of 0.0005, which was warmed up from 0.0001 over the first 50 epochs and decayed to a final learning rate of 0.00001. For the momentum contrastive learning module, we set the momentum coefficient m to 0.999, the queue size to 4096, and the temperature τ to 0.1. The loss weights α and β for the PINNE loss were both set to 1.0. An early stopping mechanism with a patience of 30 epochs was used to prevent overfitting.

Battery time serie data processing

We validated our proposed method on the widely-used NASA Ames Prognostics Center of Excellence (PCoE) battery dataset [50]. This dataset contains lifecycle data for four 18650-sized lithium-ion batteries (B0005, B0006, B0007,

and B0018) with a nominal capacity of 2 Ah. The batteries were subjected to repeated charge-discharge cycles at room temperature (24°C). The charging protocol consisted of a 1.5A Constant Current (CC) phase until the voltage reached 4.2V, followed by a Constant Voltage (CV) phase until the charge current dropped to 20mA. The discharge protocol for batteries B0005, B0006, B0007, and B0018 involved a 2A constant current load until their respective voltage cutoffs (2.7V, 2.5V, 2.2V, and 2.5V). The batteries were considered to have reached their end-of-life (EoL) when the capacity dropped by 24-30%, corresponding to an SoH of 70-76%. The number of recorded cycles for batteries B0005, B0006, B0007, and B0018 are 168, 168, 168, and 132, respectively. The state of health (SoH) was computed as the percentage ratio of the available capacity c_i to the rated nominal capacity c_i^{nominal} specific to each battery and specified by the constructor:

$$u_i = \frac{c_i}{c_i^{\text{nominal}}} \times 100 \% \quad (26)$$

Evaluation metrics

To comprehensively assess the predictive performance of our approach, figure 3, 4 we employed widely used regression metrics:

- **Mean Absolute Error (MAE):**

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{u}_i - u_i| \quad (27)$$

- **Root Mean Square Error (RMSE):**

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{u}_i - u_i)^2} \quad (28)$$

Baseline models

To provide a comprehensive and rigorous evaluation, we compared our model against a diverse set of baseline methods for SoH estimation, including our original comparisons and recent state-of-the-art models:

- **CNN [51]:** A standard data-driven approach using a Convolutional Neural Network to extract features from raw time-series data for SoH regression.
- **BPINN [51]:** A basic Physics-Informed Neural Network that incorporates physical constraints but lacks the advanced contrastive learning module of our model.
- **XGBoost-ARIMA [21]:** A hybrid model that uses XGBoost to capture complex degradation features and

ARIMA to forecast the time-series trend of the SoH, representing advanced statistical and machine learning fusion.

- **Informer [52]:** A highly efficient Transformer-based model designed for long-sequence forecasting. It uses a ProbSparse self-attention mechanism to capture significant temporal dependencies in battery data.

These baselines provided a robust benchmark, covering standard deep learning models, hybrid statistical methods, and advanced Transformer architectures.

Results and discussion

Qualitative results

We demonstrated that our augmentation approach maintained the fundamental degradation characteristics of each battery while providing sufficient diversity to enhance the robustness of the models in Fig. 5. Batteries B0006 and B0007 displayed higher initial SoH values and more gradual degradation compared to batteries B0005 and B0018, which showed more pronounced SoH degradation. These patterns were consistently reflected in both the original and augmented datasets, validating the physical consistency of our augmentation methodology.

Table 2 presented the updated MAE and RMSE values, now including the performance of the new state-of-the-art baselines. Our model demonstrated highly competitive and robust performance. Specifically, our model achieved the best performance in terms of both MAE and RMSE for batteries B0007 and B0018, recording the overall lowest MAE of 0.0764% and RMSE of 0.0994%. While the XGBoost-ARIMA model reported a slightly better MAE for B0005 and B0006, our model showed superior overall consistency and generalization, outperforming all other baselines on every metric for batteries B0007 and B0018, and achieving the best RMSE on battery B0006. For a broader comparison, the average MAE values for the baseline models were approximately 1.149% for CNN, 0.203% for BPINN, 0.327% for Informer, and 0.106% for XGBoost-ARIMA. In contrast, our model achieved a superior average MAE of 0.0952% across all four batteries Fig. 6. This demonstrated not only higher accuracy but also significantly better consistency and generalization. The reliability of our model was further highlighted by its small performance variance, with a maximum MAE difference of only 0.0420% across all batteries, showcasing its robustness to different degradation patterns.

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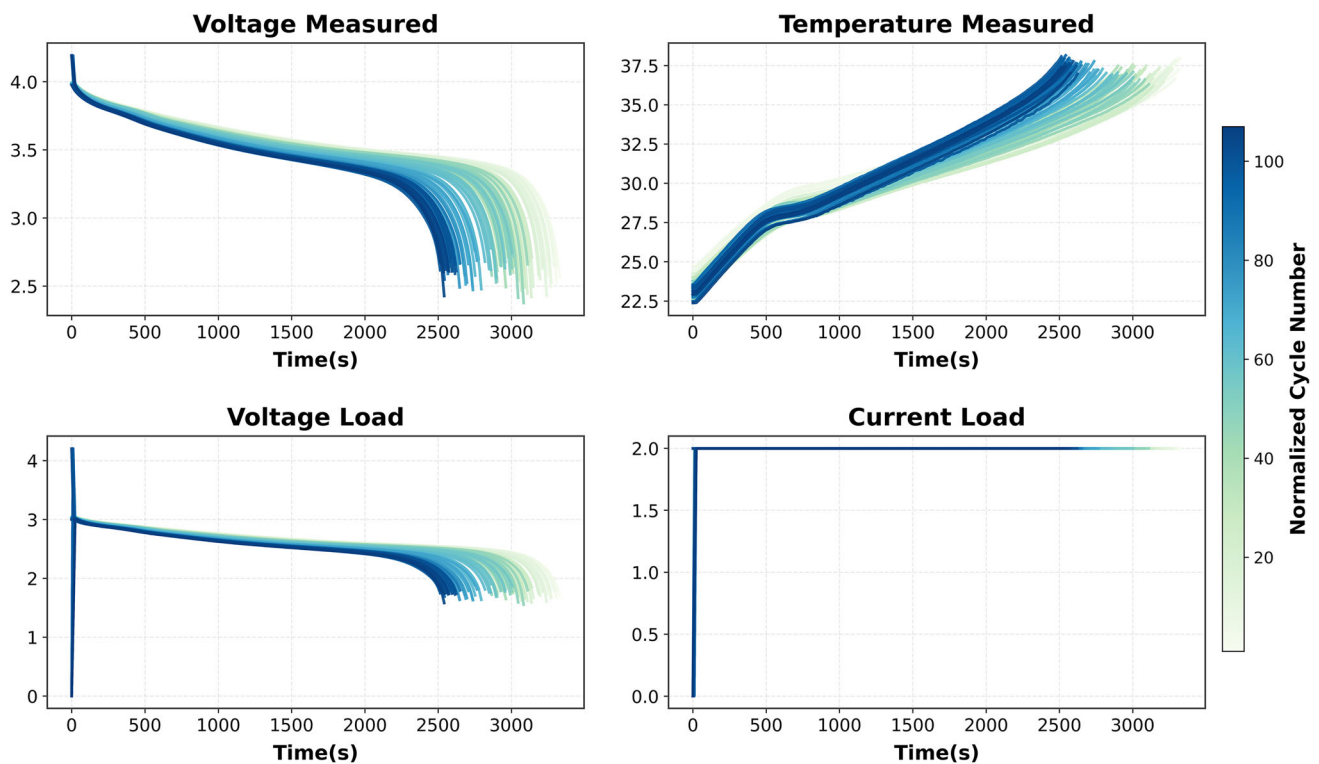


Fig. 3 B0018 battery time series data features for all cycles

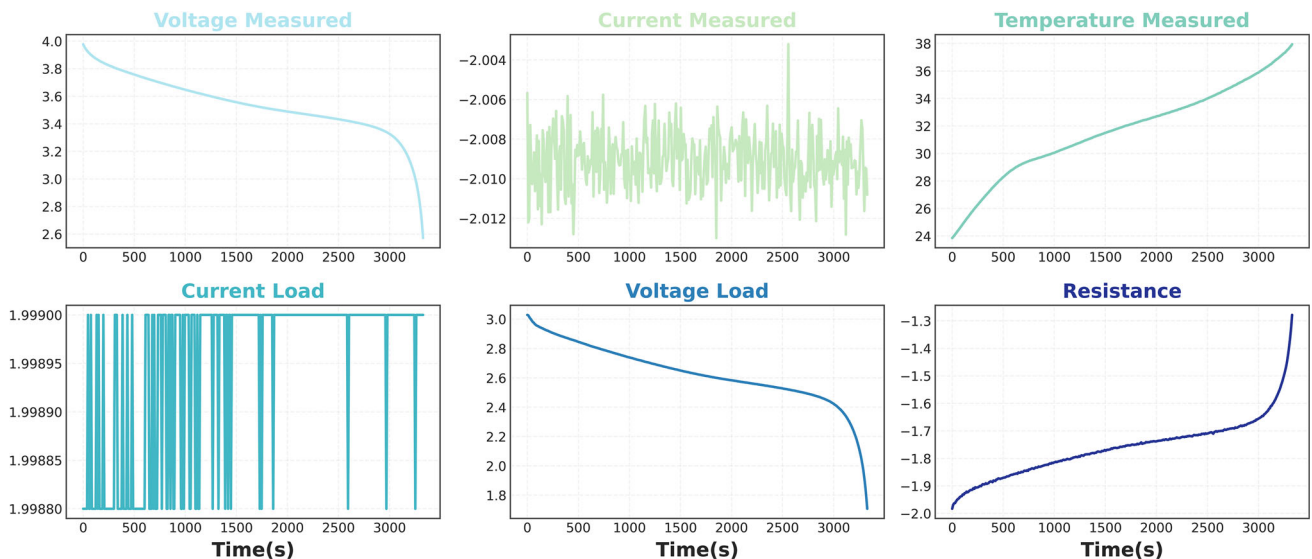


Fig. 4 B0018 battery time series data features for the first cycle

rior average MAE of 0.0952% across all four batteries. This demonstrated not only higher accuracy but also significantly better consistency and generalization. The reliability of our model was further highlighted by its small performance variance, with a maximum MAE difference of only 0.0420% across all batteries, showcasing its robustness to different degradation patterns.

Ablation study

We conducted an ablation study to assess the impact of each core component of the proposed model, as shown in Table 3. The full model achieved the lowest MAE of 0.174% and RMSE of 0.197%. Removing contrastive learning increased MAE to 0.28% and RMSE to 0.34%, indicating its critical contribution to representation learning. Exclud-

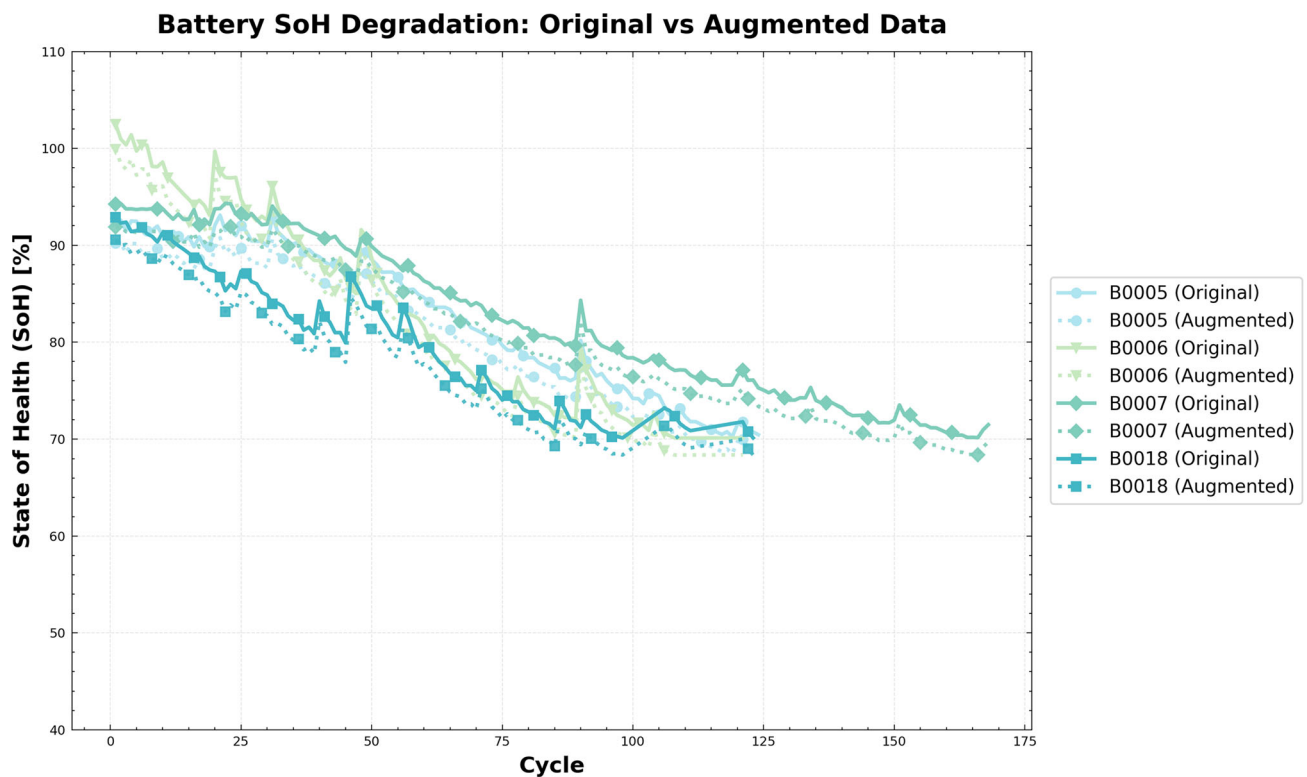


Fig. 5 Comparison of original and physically-augmented battery State of Health (SoH) degradation (B0005, B0006, B0007, and B0018). Original data (solid lines) and augmented data (dotted lines)

Table 2 Performance comparison of different models across battery datasets. The lowest value in each column is bolded

Battery		B0005	B0006	B0007	B0018
MAE (%)	CNN [51]	0.7714	1.2729	1.3952	1.1549
	BPINN [51]	0.2164	0.2351	0.1837	0.1771
	XGBoost-ARIMA [21]	0.1104	0.1025	–	–
	Informer [52]	0.2700	0.4400	0.2700	–
	Ours	0.1184	0.1080	0.0781	0.0764
RMSE (%)	CNN [51]	1.0959	1.5205	1.6129	1.4035
	BPINN [51]	0.2685	0.2937	0.2268	0.2193
	XGBoost-ARIMA [21]	0.1495	0.1591	–	–
	Informer [52]	0.3200	0.1100	0.2300	–
	Ours	0.1374	0.1080	0.1026	0.0994

ing physics constraints increased the MAE to 0.22% and RMSE to 0.28%. While removing physics-guided data augmentation raised the MAE to 0.25% and RMSE to 0.31%. Compared to the full model, omitting contrastive learning, physics-guided data augmentation, or physics constraints led to MAE increases of 61%, 44%, and 26%, respectively, and RMSE increases of 73%, 57%, and 42%, respectively. These results highlighted that all components, including contrastive learning, physics-guided data augmentation, and physics constraints, were vital for achieving optimal accuracy and robustness in State of Health estimation.

Analysis

To interpret the model, we conducted a feature importance study using Shapley values in Fig. 7, revealing that discharge capacity, voltage, resistance, and SoC were the most influential features for SoH estimation. Indeed, voltage and resistance exhibited a nearly perfect negative correlation, highlighting the electrochemical relationship where increased resistance leads to a voltage drop. Temperature is strongly correlated with both SoC and time, reflecting aging and thermal effects on degradation. Capacity mod-

Fig. 6 Battery B0005 SoH estimation results Solid blue: our model; dashed red: ground truth

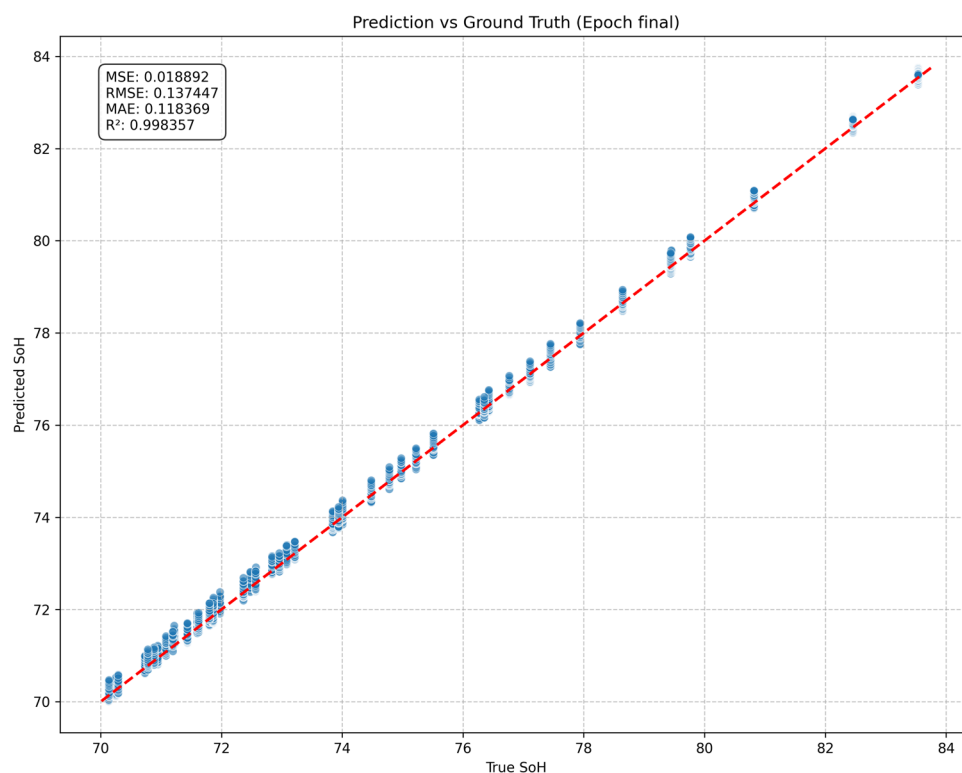


Table 3 Ablation study of the model

Variant	MAE (%)	RMSE (%)
Full Model	0.174	0.197
w/o Contrastive Learning	0.28	0.34
w/o Physics Constraints	0.22	0.28
w/o Physics-Guided data Augmentation	0.25	0.31

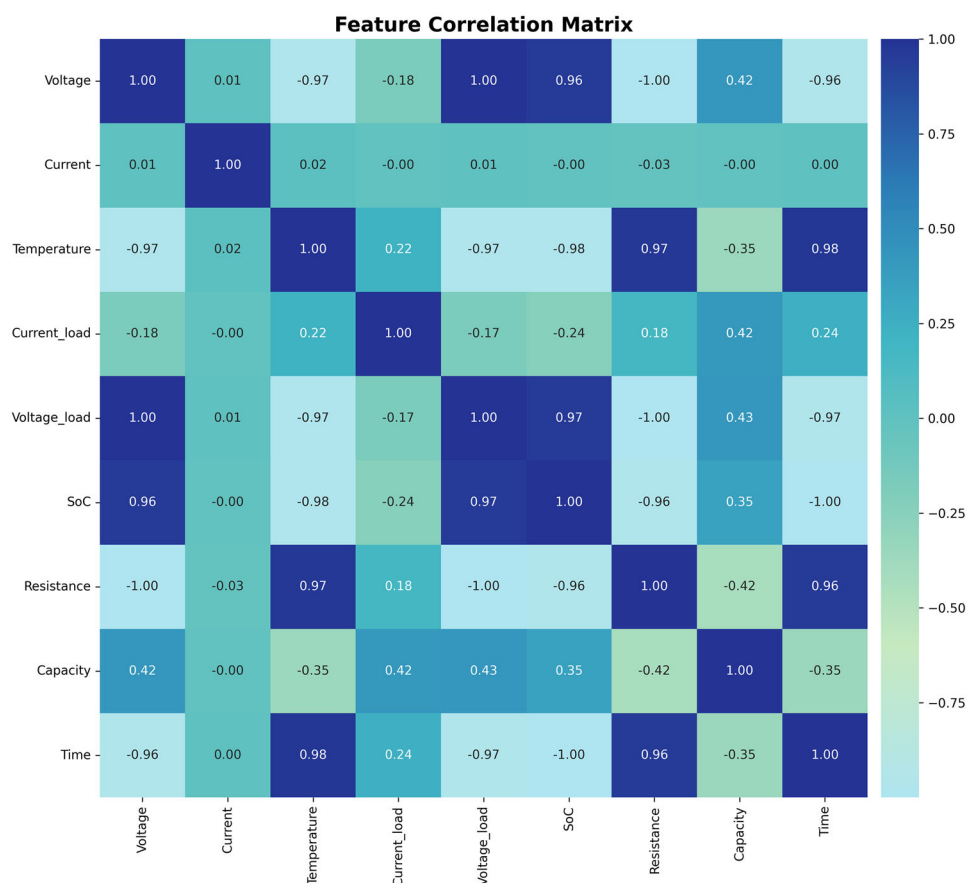
erately correlated with resistance and time. Current and current load showed minimal correlation with other features, suggesting limited direct influence. These complex interdependencies justified the use of momentum contrastive learning, which enabled the model to capture nonlinear patterns that traditional methods missed. The synergy between physics-informed modeling and contrastive learning was absolutely essential, as evidenced by the performance drop, with MAE rising from 0.174% to 0.280% and RMSE from 0.197% to 0.340% (Table 3), when contrastive learning was omitted.

In addition to estimation accuracy, the computational efficiency of the model was critical for its deployment in resource-constrained Battery Management Systems (BMS). Our proposed architecture was intentionally designed to be lightweight. With approximately 1.96 million parameters, our model was substantially smaller than large Transformer-based architectures often employed for time-series analysis. The complete training process for 500 epochs finished in approximately 2 hours and 40 minutes on a single NVIDIA GeForce RTX 4090 GPU. While the contrastive learning

approach necessitated a longer training phase, the model was highly optimized for inference. To provide a quantitative measure of its efficiency, we benchmarked the inference latency on a standard CPU. The model achieved an average single-sample latency of 0.92 s, corresponding to a throughput of over 65.2 predictions per minute. This rapid inference capability, combined with high accuracy, showed the practical application potential of our method in Battery Management Systems monitoring scenarios.

Conclusion

In this paper, we introduced a model that associates physics-informed neural network and momentum contrastive learning for accurate and robust battery State of Health (SoH) estimation. By enforcing physical constraints and leveraging advanced momentum contrastive representation learning, our model overcame key limitations of existing approaches, achieving both physical consistency and high estimation accuracy.

Fig. 7 Shapley value feature importance analysis

Our model consistently delivered low error, highlighting its strong generalization capability. Ablation studies further confirmed the critical role of each component, showing that removing contrastive learning, physics-guided data augmentation, and physical constraints increases MAE by 61%, 44%, and 26% and RMSE by 73%, 57%, and 42%, respectively. The physics-guided data augmentation strategy and the momentum contrastive learning enhance data efficiency but also ensured that estimation remains physically consistent under diverse operating conditions.

The proposed method provided a flexible and scalable approach that can be extended to different battery chemistries, operating scenarios, and degradation mechanisms. Future work will explore transfer learning across different datasets to further improve reliability and applicability. Overall, our findings demonstrate that associating physics-informed modeling with momentum contrastive learning offers a promising direction for developing accurate and reliable SoH estimation tools for battery management systems.

Author Contributions Jiwoo Jung: Methodology, Yipene Cedric Francois Bassole: Investigation. Yunsick Sung: Reviewing, Supervision.

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request. Publicly available datasets NASA battery dataset were used for training and validation purposes.

Declarations

Conflicts of Interest The authors declare no Conflict of Interest.

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